

Math 571 - Homework 6

Richard Ketchersid

Problem 6.1 (R:6:1). Suppose $\alpha : [a, b] \rightarrow \mathbb{R}$ is monotonic increasing and continuous at $x_0 \in [a, b]$. consider $f : [a, b] \rightarrow \{0, 1\}$ given by $f(x_0) = 1$ and $f(x) = 0$ for $x \neq x_0$. Show that $f \in \mathcal{R}(\alpha)$ and $\int_a^b f d\alpha = 0$.

Pick $\epsilon > 0$. Since α is continuous at x_0 take δ so that $\alpha(N_\delta(x_0)) \subseteq N_{\epsilon/2}(\alpha(x_0))$. Let $P = y_0 = a < y_1 < y_2 < y_3 = b$ where $[y_1, y_2] \subset (x_0 - \delta, x_0 + \delta)$, so that $\Delta\alpha_2 = \alpha(y_2) - \alpha(y_1) < \epsilon$. Then $M_i^{f,P} = m_i^{f,P} = 0$ for $i \neq 2$ and $M_2^{f,P} = \sup(f([y_1, y_2])) = 1$ while $m_i^{f,P} = \inf(f([y_1, y_2])) = 0$ so that

$$U(f, P) - L(F, P) = (1 - 0)\Delta\alpha_2 < \epsilon$$

Problem 6.2 (R:6:2). Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $f \geq 0$, and $\int_a^b f dx = 0$, then $f = 0$.

Note that where Rudin asks you to compare with (R:6:1). You might think that these are not comparable since (R:6:1) is about $\mathcal{R}(\alpha)$ while (R:6:2) is about $\mathcal{R}$, but taking $\alpha = \text{id}$ in (R:6:1) allows you to make the comparison.

This is really almost trivial. If $f \neq 0$, then $f(x) > 0$ for some $x \in [a, b]$, but then $f(x) > \delta > 0$ and so there is an open nbhd of x , $N_\delta(x) = (x - \delta, x + \delta)$ so that $f((x - \delta, x + \delta) \cap [a, b]) \subset (\delta, \infty)$. Say $(c, d) \subseteq (x - \delta, x + \delta) \cap [a, b]$, then clearly $\int_a^b f dx \geq \delta(d - c) > 0$.

The difference between the examples from (R:6:1) and (R:6:2) is that in the former, the function is not continuous. In fact $\int_a^b f dx = 0$ whenever $\{x \in [a, b] \mid f(x) \neq 0\}$ has **measure 0**. A set Z has measure 0 whenever

$$0 = \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) \mid Z \subseteq \bigcup_{i=1}^{\infty} (a_i, b_i) \right\}$$

Problem 6.3 (R:6.3). Define $\beta_i : [-1, 1] \rightarrow [0, 1]$ by $\beta_i = 0$ for $x < 0$ and $\beta_i = 1$ for $x > 0$, then $\beta_1(0) = 0$, $\beta_2(0) = 1$, and $\beta_3(0) = 1/2$. In particular β_i has a simple discontinuity at 0 with $\beta_1(0-) = \beta_1(0) = 0$ (continuous from the left), $\beta_2(0+) = \beta_2(0) = 1$ (continuous from the right), while β_3 is neither continuous from the left or right. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be bounded. show that

- (a) $f \in \mathcal{R}(\beta_1)$ iff $f(0+) = f(0)$, that is, f is continuous from the right at 0.
- (b) $f \in \mathcal{R}(\beta_2)$ iff $f(0-) = f(0)$, that is, f is continuous from the left at 0.
- (c) $f \in \mathcal{R}(\beta_3)$ iff f is continuous at 0.

- (d) Given an interval $[a, b]$ and a point $c \in [a, b]$, let $\beta_{j,a,b,c}$ ($j = 1, 2$) be the functions on $[a, b]$ which equal 0 on $[a, c)$, 1 on $(c, b]$, and are 0 or 1 at c depending on whether j is 1 or 2. (So for $j = 1, 2$, Rudin's β_j is $\beta_{j,-1,1,0}$ in this notation.) **State the results analogous to Rudin's (a) and (b) for integrals $\int_a^b f(x) d\beta_{j,a,b,c}(x)$.** Do not hand in proofs. But think things through carefully; you will be graded on the correctness of your answers. The cases $c = a$ and $c = b$ are slightly different from the general case, so I recommend that you first state the results for $c \in (a, b)$, then say how they must be modified for those cases.
- (e) Assuming the results of part (e), deduce that a function f on an interval $[a, b]$ is integrable with respect to *every* increasing function α if and only if it is continuous.

(a) - (c): These are all very similar. For the "if" direction ($\Leftarrow$). It suffices to consider partitions that include 0 so that

$$P : -1 = x_0 < x_1 < \dots < x_k = 0 < \dots < x_n = 1$$

where $x_k = 0$. For β_i we have

$$(\Delta\beta_i)_k = \beta_i(0) - \beta_i(x_{k-1}) = \begin{cases} 0 & i = 1 \\ 1 & i = 2 \\ 1/2 & i = 3 \end{cases}$$

and

$$(\Delta\beta_i)_{k+1} = \beta_i(x_{k+1}) - \beta_i(0) = \begin{cases} 1 & i = 1 \\ 0 & i = 2 \\ 1/2 & i = 3 \end{cases}$$

All other $(\Delta\beta_i)_j = 0$ and thus we see

$$\begin{aligned} U(f, P) - L(f, P) &= (M_k - m_k)(\Delta\beta_i)_k + (M_{k+1} - m_{k+1})(\Delta\beta_i)_{k+1} \\ &= \begin{cases} M_{k+1} - m_{k+1} & i = 1 \\ M_k - m_k & i = 2 \\ \frac{1}{2}((M_k - m_k) + (M_{k+1} - m_{k+1})) & i = 3 \end{cases} \end{aligned}$$

Now $f \in \mathcal{R}(\beta_i)$ iff for all $\epsilon > 0$ there is a P so that

$$U(f, P) - L(f, P) < \epsilon \iff \begin{cases} M_{k+1} - m_{k+1} < \epsilon & i = 1 \\ M_k - m_k < \epsilon & i = 2 \\ \frac{1}{2}((M_k - m_k) + (M_{k+1} - m_{k+1})) < \epsilon & i = 3 \end{cases}$$

Take the $i = 1$ case, this says that for all $\epsilon > 0$ there is $x_{k+1} = h > 0$ so that $\sup(f([0, h]) - \inf(f([0, h])) < \epsilon$ which says exactly that $f(0+) = f(0)$. Similarly for $i = 2$ and $i = 3$.

The "only if" ($\Rightarrow$) is as follows. Suppose P is a partition so that $U(f, P, \beta_i) - L(f, P, \beta_i) < \epsilon$. Without loss of generality we may assume 0 is in the partition so that $x_k = 0$ and there are

pieces $[x_{k-1}, 0]$ and $[0, x_{k+1}]$ Then you have exactly the same situation as above and get:

$$U(f, P) - L(f, P) < \epsilon \iff \begin{cases} M_{k+1} - m_{k+1} < \epsilon & i = 1 \\ M_k - m_k < \epsilon & i = 2 \\ \frac{1}{2}((M_k - m_k) + (M_{k+1} - m_{k+1})) < \epsilon & i = 3 \end{cases}$$

Thus, it is clear that, say, in the case of $i = 1$, that $\sup(f[0, x_{k+1}]) - \inf(f([0, x_{k+1}]))$ can be made arbitrarily small and thus $f(0+) = f(0)$. Similarly, for the other cases.

(d): Assume $c \in (a, b)$, then the exact same argument as above for (a) and (b) works. So if $f(c+) = f(c)$, then $f \in \mathcal{R}(\beta_{1,a,b,c})$ and if $f(c-) = f(c)$, then $f \in \mathcal{R}(\beta_{2,a,b,c})$ and the point is to look at a partition $P = a < c - h < c < c + h < b$, then

$$U(f, \beta_{i,a,b,c}, P) = \begin{cases} \sup f([c, c+h]) - \inf f([c, c+h]) & \text{if } i = 1 \\ \sup f([c-h, c]) - \inf f([c-h, c]) & \text{if } i = 2 \end{cases}$$

If $c \in \{a, b\}$, then $f(a+) = f(a) \iff f \in \mathcal{R}(\beta_{1,a,b,a})$ and $f(b-) = f(b) \iff f \in \mathcal{R}(\beta_{2,a,b,b})$.

(e): If $f \in \mathcal{R}(\beta_{i,a,b,c})$ for all $c \in [a, b]$ and $i \in \{1, 2\}$, then $f(c) = f(c+) = f(c-)$ for all $c \in (a, b)$ and $f(a+) = f(a)$ and $f(b-) = f(b)$ so f must be continuous on $[a, b]$.

Problem 6.4 (B:3.3:3 – An α which does all its increasing on the Cantor set.). Let $\alpha : [0, 1] \rightarrow [0, 1]$ be the Cantor function from Problem 4.10 of Homework 4. The main fact about α is that it is an increasing (non-decreasing) continuous function from $[0, 1]$ onto $[0, 1]$ that is absolutely flat off the Cantor set. So α only increases on a measure 0 set, yet is surjective!

- Show that if f is a continuous real-valued function on $[0, 1]$ which is zero on all points of the Cantor set, then $\int f d\alpha = 0$.
- Give an example of a function f as in (a) which is not the zero function.
- Deduce from (a) that if f and g are two continuous real-valued functions on $[0, 1]$ which agree on all points of the Cantor set, then $\int f d\alpha = \int g d\alpha$.
- Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function such that $f(x) = 0$ for all x in the Cantor set, and $f(x) = 1$ for all other x . Show that $f \notin \mathcal{R}(\alpha)$.
- Why does the result of (d) not contradict the result of (a) above? Why does it not contradict the result of **R:6.6**?

(a): Fix $\epsilon > 0$. f is continuous and hence uniformly continuous, so get $\delta > 0$ so that $|f(x) - f(y)| < \epsilon$ for all $|x - y| < \delta$, where $|f(x)| < M$ for all $x \in [0, 1]$. Let $P = \{x_0, \dots, x_n\}$ be a partition with $\Delta x_i < \delta$. Then

$$U(f, \alpha, P) = \sum_i \sup f([x_{i-1}, x_i]) \Delta \alpha_i \leq \sum_{i \in A} \epsilon \Delta \alpha_i \leq \epsilon$$

where $A = \{i \mid [x_{i-1}, x_i] \cap \mathcal{C} \neq \emptyset\}$.

Similarly, $L(f, \alpha, P) \geq -\epsilon$. So we get $U(f, \alpha) = 0 = L(f, \alpha)$.

(b): Build $f_n \rightarrow f$ uniformly as follows. Let $\mathcal{C}_i$ be the closed sets built in constructing the Cantor set. Let f_{n+1} be a piecewise linear function determined by f_n as follows. For a maximal segment $[a, b] \subset \mathcal{C}_n$, we connect $(a + \frac{b-a}{3}, 0)$, $(\frac{b+a}{2}, 2^{-(n+1)})$, and $(b - \frac{b-a}{3}, 0)$ with straight lines. So clearly $\sup\{|f_{n+1}(x) - f_n(x)| \leq 2^{-(n+1)}\}$ and $f_{n+1}(x) = 0 \iff x \in \mathcal{C}_{n+1}$. (See [here](#)). Each f_n is continuous and $f_n \rightarrow f$ uniformly so f is continuous and clearly non-zero.

(c): This is clear since $\int (f - g) d\alpha = \int f d\alpha - \int g d\alpha$.

(d): Let f be as defined. Let P be a partition, then

$$U(f, \alpha, P) - L(f, \alpha, P) = \sum_{i \in A} \Delta\alpha_i = 1$$

where $i \in A \iff \mathcal{C} \cap [x_{i-1}, x_i] \neq \emptyset$. The point is $\Delta\alpha_i = 0$ for $i \notin A$. So $f \notin \mathcal{R}(\alpha)$.

(e): This does not contradict (a) as f is not continuous, $D(f) = \{x \mid f \text{ is discontinuous at } x\} = \mathcal{C}$.

Problem 6.5 (B:6.4:9 – Functions with only countable many discontinuities are integrable.). Let f be bounded on $[a, b]$ with at most countable many discontinuities on $[a, b]$. Let $\alpha : [a, b] \rightarrow \mathbb{R}$ is monotonic increasing and α is continuous at every discontinuity of f . Show that $f \in \mathcal{R}(\alpha)$.

Hint: Fix an enumeration $S = \{s_i \mid i \in \mathbb{N}\}$ of the discontinuities of f . Fix $\epsilon > 0$ and $\epsilon_i > 0$ so that $\sum_i \epsilon_i \leq \epsilon$. Since α is continuous at s_i fix δ_i so that $\alpha(N_{\delta_i}(s_i)) \subset N_{\epsilon_i}(\alpha(s_i))$. For $x \notin S$, fix δ_x so that $f(N_{\delta_x}(x)) \subset N_\epsilon(f(x))$. Now $\mathcal{O} = \{N_{\delta_i}(s_i) \mid i \in \mathbb{N}\} \cup \{N_{\delta_x}(x) \mid x \notin S\}$ is an open cover of $[a, b]$. Apply compactness to get a finite subcover and then do something *similar* (not the same) as in the proof of 6.10.

Proof 1: Fix $\epsilon > 0$ and $\epsilon_i > 0$ so that $\sum_i \epsilon_i < \epsilon$. Let $S = \{s_i \mid i \in \mathbb{N}\}$ be the discontinuities of f . Since α is continuous at s_i let δ_i be so that $\alpha(N_{\delta_i}(s_i)) \subset N_{\epsilon_i}(\alpha(s_i))$. For $x \notin S$ let δ_x be chosen so that $f(N_{\delta_x}(x)) \subseteq N_\epsilon(f(x))$. Let $\mathcal{O} = \{N_{\delta_i}(s_i) \mid i \in \mathbb{N}\} \cup \{N_{\delta_x}(x) \mid x \notin S\}$ be the associated open cover of $[a, b]$. Let $\mathcal{O}' \subseteq \mathcal{O}$ be a finite subcover of $[a, b]$. Notice that $\mathcal{O}'$ consists of intervals (u_i, v_i) and we may assume that $a = u_0 < u_1 < v_0 < u_2 < v_1 < u_3 < v_2 \cdots$ (a “chain”). Thus we define $x_0 = a < x_1 = (u_1 + v_0)/2 < x_2 = (u_2 + v_1)/2 < x_{n-1} = (u_{n-1} + v_{n-2})/2 < x_n = v_n = b$. Thus $[x_{i-1}, x_i] \subset N_{\delta_j}(s_j)$ for some j or $[x_{i-1}, x_i] \subset N_{\delta_x}(x)$ for some $x \notin S$.

Let $T = \{i \mid [x_{i-1}, x_i] \subset N_{\delta_i}(s_i) \text{ for some } s_i \in S\}$. Then letting $|f|(x) \leq M$ and $\alpha(b) - \alpha(a) = N$:

$$\begin{aligned} \sum_{i=1}^n |M_i - m_i|(\alpha(x_i) - \alpha(x_{i-1})) &= \sum_{i \in T} |M_i - m_i|(\alpha(x_i) - \alpha(x_{i-1})) + \sum_{i \notin T} |M_i - m_i|(\alpha(x_i) - \alpha(x_{i-1})) \\ &< \sum_{i \in T} 2M\epsilon_i + \sum_{i \notin T} \epsilon(\alpha(x_i) - \alpha(x_{i-1})) \\ &\leq 2M\epsilon + N\epsilon = \epsilon(2M + N) \end{aligned}$$

Proof 2: (The following seems to be an option that I see commonly, but not carried out correctly. I thought I would write it out correctly here.)

Start like the above. Since α is continuous at s_i pick (a_i, b_i) satisfying:

- $(a_i, b_i) \cap (a_j, b_j) = \emptyset$. (mutually disjoint)
- $s_i \in (a_i, b_i)$.
- $\alpha((a_i, b_i)) \subseteq S_{\epsilon/2}(s_i)$ so that if $t, t' \in (a_i, b_i)$, then $|\alpha(t') - \alpha(t)| < \epsilon_i$. Where $\sum_i \epsilon_i = \epsilon$ and ϵ_i will be chosen at the end.

Let $K = [a, b] - \bigcup_i (b_i, a_i)$. K is closed and bounded, hence compact. Since f is continuous on K it is uniformly continuous and thus we can pick $\delta > 0$ so that $|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$ for all $x, y \in K$.

$\mathcal{O} = \{(x - \delta/2, x + \delta/2) \mid x \in K\} \cup \{(a_i, b_i) \mid i \in \mathbb{N}\}$ is an open cover of $[a, b]$ and hence has a finite subcover $\mathcal{O}'$. Let $\mathcal{O}' = \{(u_i, v_i) \mid i < m\}$ we may assume that for no $i \neq j$ do we have $(u_i, v_i) \subset (u_j, v_j)$, as we could just toss out (u_i, v_i) in this case. So $u_0 < a < u_1 < v_0 < u_2 < v_1 < \dots < u_{m-1} < v_{m-2} < b < v_{m-1}$. For $i = 1, \dots, m-2$ let $y_i = (u_i + v_{i-1})/2$ and set $y_0 = a$ and $y_{m-1} = b$ and let $P = \{y_i \mid i = 0, \dots, m-1\}$. Then we know for each $i = 1, \dots, m-1$ that either $[u_{i-1}, u_i] \subset (a_j, b_j)$ for some j or else $[u_{i-1}, u_i] \subset (x - \delta/2, x + \delta/2)$ for some $x \in K$.

Let $A = \{i \mid [u_{i-1}, u_i] \subset (a_{j_i}, b_{j_i}) \text{ for some } j_i\}$, then for $i \in A$ we have $\Delta\alpha_i = \alpha(u_i) - \alpha(u_{i-1}) < \epsilon_{j_i}$ and for $i \notin A$, $|M_i^{f,P} - m_i^{f,P}| < \epsilon$. Thus

$$\begin{aligned} U(f, P, \alpha) - L(f, P, \alpha) &= \sum_{i=1}^{m-1} |M_i^{f,P} - m_i^{f,P}| \Delta\alpha_i \\ &\leq \sum_{i \in A} |M_i^{f,P} - m_i^{f,P}| \epsilon_{j_i} + \sum_{i \notin A} \epsilon \Delta\alpha_i \leq 2M\epsilon + \epsilon(\alpha(b) - \alpha(a)) \end{aligned}$$

where $M = \sup |f(x)|_{x \in [a, b]}$.

Since M and $\alpha(b) - \alpha(a)$ are fixed constants we can make the $\epsilon(2M + \alpha(b) - \alpha(a))$ arbitrarily small. Thus $f \in \mathcal{R}(\alpha)$.

Problem 6.6 (R:6:6 or B:6.4:10 – A Riemann Integrable function with uncountably many discontinuities.). Let $f : [0, 1] \rightarrow \mathbb{R}$ be bounded and continuous off of the Cantor set $\mathcal{C}$. Show that $f \in \mathcal{R}$.

Recall the construction of the Cantor set. $C_0 = [0, 1]$ $C_1 = [0, 1] - (1/3, 2/3)$ (removing middle third). $C_2 = C_1 - (1/9, 2/9) - (7/9, 8/9)$, again remove middle thirds from what was left.

Notice the lengths of what is removed: $1/3$, $1/3 + 2/9$, $1/3 + 2/9 + 4/27$, etc. Consider

$$\sum_{i=0}^{\infty} \frac{2^i}{3^{i+1}} = \frac{1}{3} \sum_{i=0}^{\infty} \left(\frac{2}{3}\right)^i = \frac{1}{3} \left(\frac{1}{1 - 2/3}\right) = 1$$

We can cover C_i by 2^i many disjoint intervals of length $(1/3)^i + \epsilon$ for any ϵ . Since $\mathcal{C} = \bigcap C_i$ we see that $\mathcal{C}$ has measure 0 as defined above.

Suppose f is continuous off of a measure 0 set $Z \subset [a, b]$. Let $\mathcal{O}$ be an open cover of Z by intervals (a_i, b_i) so that $\sum_i (b_i - a_i) < \epsilon$. For each $x \notin Z$ take δ_x so that $f(N_{\delta_x}(x)) \subset N_{\epsilon/2}(f(x))$. As $[a, b]$ is compact we can find a finite subcover $\{(u_i, v_i) \mid i = 1, \dots, n\}$ so that $u_1 < a < u_2 < v_1 < u_3 < v_2 < \dots < u_n < v_{n-1} < b < v_n$ where each (u_i, v_i) is from our cover of Z or else is one of the $N_{\delta_x}(x)$.

Use $x_0 = a$, $x_i = (u_{i+1} + v_i)/2$ for $i < n$, and $x_n = b$ as the partition: $P = a = x_0 < x_1 < \dots < b = x_n$. Let M be a bound on $|f|$ on $[a, b]$. Let $T = \{i \mid (x_{i-1}, x_i) \subseteq (a_j, b_j) \text{ for some } j\}$. Then we have

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{i \in T} (M_i - m_i) \Delta x_i + \sum_{i \notin T} (M_i - m_i) \Delta x_i \\ &< \sum_{i \in T} 2M \Delta x_i + \sum_{i \notin T} \epsilon \Delta x_i \\ &\leq 2M\epsilon + \epsilon(b - a) = \epsilon(2M + (b - a)) \end{aligned}$$

Summary of Problems 6.4 - 6.6 and a generalization: Let $\text{Dis}(f) = \{x \mid x \text{ is discontinuous at } x\}$. Let $\alpha : [0, 1] \rightarrow [0, 1]$ be the Cantor function (“*devil’s staircase*”) and let $f = \chi_{\mathcal{C}}$ be the characteristic function of the Cantor set, $f(x) = 1$ for $x \in \mathcal{C}$ and $f(x) = 0$ for $x \notin \mathcal{C}$. Let $g : [0, 1] \rightarrow \mathbb{R}$ be any bounded function where $\text{Dis}(g) \cap \text{Dis}(\alpha) = \emptyset$.

1. If $\text{Dis}(g)$ is countable, then $g \in \mathcal{R}(\alpha)$.
2. $f \in \mathcal{R}$, but $f \notin \mathcal{R}(\alpha)$.

Is there a general pattern? Define $A \subseteq [a, b]$ *has measure 0* to mean

$$\inf \left\{ \sum_n (b_n - a_n) \mid A \subseteq \bigcup_n (a_n, b_n) \right\} = 0$$

In words: We can cover A by a collection of intervals, the sum of whose lengths can be made arbitrarily small. With this definition, we have:

Theorem 1. If $f : [a, b] \rightarrow \mathbb{R}$ is bounded, then

$$\text{Dis}(f) \text{ has measure } 0 \implies f \in \mathcal{R}$$

This covers our first example as *any countable subset of $[a, b]$ has measure 0*.

We can extend this notion to increasing $\alpha : [a, b] \rightarrow \mathbb{R}$. Say $A \subseteq [a, b]$ *has α -measure 0* if

$$\inf \left\{ \sum_n (\alpha(b_n) - \alpha(a_n)) \mid A \subseteq \bigcup_n (a_n, b_n) \right\} = 0$$

This is the same as measure 0 when α is the identity function. Then we have:

Theorem 2. If $\alpha : [a, b] \rightarrow \mathbb{R}$ is monotonically increasing and $f : [a, b] \rightarrow \mathbb{R}$ is bounded and $\text{Dis}(f) \cap \text{Dis}(\alpha) = \emptyset$, then

$$\text{Dis}(f) \text{ has } \alpha\text{-measure } 0 \implies f \in \mathcal{R}(\alpha)$$

This covers our other two examples. In particular, if α is the Cantor function, then $[0, 1] \setminus \mathcal{C}$ has α -measure 0, or, said differently, $\mathcal{C}$ has α -measure 1.

So any bounded function continuous off the Cantor set is Riemann integrable since the Cantor set has measure 0, but if α is the Cantor function, then the Cantor set has α -measure > 0 and hence f might no longer be in $\mathcal{R}(\alpha)$.

Problem 6.7 (B:6.4:8 – Integrability piece by piece.). Let $s_1 < s_2 < \dots < s_n < \dots$ be elements of an interval $[a, b]$ such that $s_1 = a$ and $\lim_{n \rightarrow \infty} s_n = b$. Let α be an increasing function on $[a, b]$, and let f be any bounded real-valued function on $[a, b]$. For each positive integer i , let f_i denote the restriction of f to $[s_i, s_{i+1}]$, and α_i the restriction of α to that interval.

Note: There is an error in this problem. You need to add that either $f(b-) = f(b) = 0$ or that $\alpha(b-) = \alpha(b)$.

Prove that $f \in \mathcal{R}(\alpha)$ if and only if for every i , $f_i \in \mathcal{R}(\alpha_i)$, and that when this holds, we have

$$\int_a^b f \, d\alpha = \sum_{i=1}^{\infty} \left(\int_{s_i}^{s_{i+1}} f \, d\alpha \right),$$

with the sum on the right absolutely convergent. (The i th summand of that sum could, of course, be written more formally as $\int_{s_i}^{s_{i+1}} f_i \, d\alpha_i$.)

Bonus: Find an example where this is false if the extra assumptions is not made.

Suppose $f \in \mathcal{R}(\alpha)$, combining Theorem 6.12 (c) and Theorem 6.13 we have:

$$\begin{aligned} \left| \int_a^b f \, d\alpha \right| &= \left| \sum_{i=1}^n \int_{s_i}^{s_{i+1}} f \, d\alpha + \int_{s_n}^b f \, d\alpha \right| \\ &\leq \sum_{i=1}^n \left| \int_{s_i}^{s_{i+1}} f \, d\alpha \right| + \left| \int_{s_n}^b f \, d\alpha \right| \\ &\leq \sum_{i=1}^n \int_{s_i}^{s_{i+1}} |f| \, d\alpha + \int_{s_n}^b |f| \, d\alpha \\ &= \int_a^b |f| \, d\alpha \end{aligned}$$

So all we really need to do is to show $\lim_{t \rightarrow b-} \int_t^b |f| \, d\alpha \rightarrow 0$. But here we have $\int_t^b |f| \, d\alpha \leq \sup\{|f(x)| \mid x \in [t, b]\}(\alpha(b) - \alpha(t)) = M_t \Delta\alpha_t$. This is where we need the extra assumption that was missing from Bergman's notes.

In the other direction, assume $f_i \in \mathcal{R}(\alpha)$ and $\sum_{i=1}^{\infty} \left(\int_{s_i}^{s_{i+1}} f \, d\alpha \right)$ is absolutely convergent. Fix $\epsilon > 0$ and get $M_{s_n} \Delta\alpha_{s_n} < \frac{\epsilon}{2}$, again using our additional hypotheses. Find partitions P_i for each $[s_{i-1}, s_i]$ so that $U(f_i, \alpha, P_i) - L(F_i, \alpha, P_i) < \frac{\epsilon}{2^{i+1}}$, then let $P = \bigcup P_i \cup \{[s_n, b]\}$ so that

$$\begin{aligned} U(f, \alpha, P) - L(f, \alpha, P) &= \sum_{i=1}^n U((f_i, \alpha, P_i) - L(F_i, \alpha, P_i)) \\ &\quad + \left(\sum f([s_n, b]) - \inf f([s_n, b]) \right) (\alpha(s_n) - \alpha(b)) \\ &< \sum_{i=1}^n \frac{\epsilon}{2^{i+1}} + \frac{\epsilon}{2} < \epsilon \end{aligned}$$

Bonus. We already have such an example in Problem 6.3 with $\beta_{1,a,b,b}$ and say where $f(x) = 0$ on $[a, b)$ and $f(b) = 1$. Then $f \notin \mathcal{R}(\alpha)$ and $\int_{s_n}^{s_{n+1}} f \, d\alpha = 0$.

Problem 6.8 (B:6.5:2 – Strengthening of 6.16). Show that Theorem 6.16 (p.130) remains valid if the assumption that f is continuous on $[a, b]$ is replaced by the condition that it is bounded, and is continuous at each of the points s_n . Specifically -

We see that no continuity condition is needed until display (24). Say why that display is valid under the condition stated above. Then modify the remainder of the proof (in which Rudin uses integrability of f with respect to α_2 , which under his assumptions follows from Theorem 6.8) to conclude that (23) holds with the upper, respectively the lower integral in place of the left-hand side. Conclude that $f \in \mathcal{R}(\alpha)$ and that (23) holds.

Problem 6.9 (B:6.6:1. – A sort of derivative formula for Riemann-Stieltjes). On $[a, b]$, let α be a strictly increasing function (as defined in **B:5.2:1 (b)**) and f a continuous function, and for $x \in [a, b]$ define $F(x) = \int_a^x f(t) d\alpha(t)$.

Show that for all $x \in [a, b]$,

$$\frac{dF(x)}{d\alpha(x)} = f(x),$$

where the left-hand side is defined as $\lim_{t \rightarrow x} (F(x) - F(t)) / (\alpha(x) - \alpha(t))$, and the equality includes the assertion that this limit exists.

First take $t < x$ and notice by Theorem 6.12

$$m_t(\alpha(x) - \alpha(t)) \leq F(x) - F(t) = \int_t^x f(t) d\alpha(t) < M_t(\alpha(x) - \alpha(t))$$

From this, we get the following using the fact that $\alpha(x) > \alpha(t)$, i.e., α is *strictly increasing*:

$$m_t \leq \frac{F(x) - F(t)}{\alpha(x) - \alpha(t)} = \frac{\int_t^x f(t) d\alpha(t)}{\alpha(x) - \alpha(t)} < M_t$$

where $m_t = \inf f([t, x])$ and $M_t = \sup f([t, x])$. As f is continuous at x , $\lim_{t \rightarrow x^-} m_t = f(x) = \lim_{t \rightarrow x^-} M_t$ and so, essentially by the *Squeeze Theorem*, $\lim_{t \rightarrow x^-} \frac{F(x) - F(t)}{\alpha(x) - \alpha(t)} = f(x)$.

The argument is essentially the same if $x < t$.

Note: This really is just the usual argument for the FTC for Riemann Integration.

Problem 6.10 (B:6.6:5 – Repeated Riemann-Stieltjes integration.). We shall obtain here a generalization of the result of **B:6.6:2** in which ds and dt are replaced by more general expressions $d\alpha(s)$ and $d\beta(t)$.

- (a) Suppose α, β are increasing functions on $[a, b]$ such that $\alpha \in \mathcal{R}(\beta)$, and $f \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$. Show that $\int_a^x f(t) d\alpha(t)$, regarded as a function of x , belongs to $\mathcal{R}(\beta)$, and that

$$\int_{t=a}^b \left(\int_{s=a}^t f(s) d\alpha(s) \right) d\beta(t) = \int_{t=a}^b (\beta(b) - \beta(t)) f(t) d\alpha(t).$$

(Suggestion: First consider the case where f is everywhere > 0 since in this case one can easily describe the least and greatest values of $\int_{s=a}^t f(s) d\alpha(s)$ on any interval as the values at the respective ends of the interval; then get the general case by writing any $f \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$ as a difference of two nonnegative-valued functions in $\mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$. Cf. the method of passing from part **B:6.5:1 (b)** to (c).)

- (b) Show that given continuous functions u and v on $[a, b]$, there exists a continuous function w of two variables such that for any $f \in \mathcal{R}$ on $[a, b]$, one has

$$\int_{t=a}^x u(t) \left(\int_{s=a}^t v(s) f(s) ds \right) dt = \int_{t=a}^x w(x, t) f(t) dt.$$

Hint: Prove the case where u and v are nonnegative-valued using (a).

(a) Following the hint, first assume $f(x) \geq 0$ for all x . Since α is increasing and $f \geq 0$, $F(t) = \int_a^t f d\alpha$ is a monotonically increasing function. Because f is bounded, $0 \leq f(x) \leq M$ for some $M > 0$, meaning for $x < y$, $0 \leq F(y) - F(x) \leq M(\alpha(y) - \alpha(x))$.

We are given $\alpha \in \mathcal{R}(\beta)$, which is equivalent to the condition that for any $\epsilon > 0$, there exists a partition $P = \{x_0, \dots, x_n\}$ such that $\sum \Delta\alpha_i \Delta\beta_i < \epsilon$. For such a partition, the upper and lower sums for F with respect to β satisfy:

$$U(P, F, \beta) - L(P, F, \beta) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) \Delta\beta_i \leq M \sum_{i=1}^n \Delta\alpha_i \Delta\beta_i < M\epsilon.$$

Thus, $F \in \mathcal{R}(\beta)$.

To prove the identity, let P be such a partition. Consider the sum $\sum_{i=1}^n F(x_i) \Delta\beta_i$, an upper sum for $\int_a^b F d\beta$. We can write $F(x_i) = \sum_{j=1}^i \Delta F_j$, where $\Delta F_j = F(x_j) - F(x_{j-1}) = \int_{x_{j-1}}^{x_j} f d\alpha$. By changing the order of summation:

$$\sum_{i=1}^n F(x_i) \Delta\beta_i = \sum_{i=1}^n \left(\sum_{j=1}^i \Delta F_j \right) \Delta\beta_i = \sum_{j=1}^n \Delta F_j \left(\sum_{i=j}^n \Delta\beta_i \right) = \sum_{j=1}^n \Delta F_j (\beta(b) - \beta(x_{j-1})).$$

Similarly, the lower sum is:

$$\sum_{i=1}^n F(x_{i-1}) \Delta\beta_i = \sum_{j=1}^n \Delta F_j (\beta(b) - \beta(x_j)).$$

Define $G(t) = (\beta(b) - \beta(t))f(t)$. Since $f \in \mathcal{R}(\alpha)$ and $\beta \in \mathcal{R}(\alpha)$ (since $\alpha \in \mathcal{R}(\beta)$), $G \in \mathcal{R}(\alpha)$. The integral $\int_a^b G d\alpha = \sum_{j=1}^n \int_{x_{j-1}}^{x_j} (\beta(b) - \beta(t))f(t) d\alpha(t)$ is strictly bounded between the two partial sums above because $\beta(b) - \beta(x_j) \leq \beta(b) - \beta(t) \leq \beta(b) - \beta(x_{j-1})$ on $[x_{j-1}, x_j]$ and $f \geq 0$. Since the upper and lower sums differ by at most $M\epsilon$, both $\int_a^b F d\beta$ and $\int_a^b G d\alpha$ lie in the same interval of length $< M\epsilon$. Since ϵ is arbitrary, they must be equal.

For a general $f \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$, since f is bounded, choose $M > 0$ such that $f(x) + M \geq 0$. Let $f_1 = f + M$ and $f_2 = M$. Both are non-negative and belong to $\mathcal{R}(\alpha) \cap \mathcal{R}(\beta)$. By the argument above, the identity holds for f_1 and f_2 . By linearity of the integral, it holds for $f = f_1 - f_2$.

(b) First assume $u, v \geq 0$. Let $\beta(t) = \int_a^t u(r) dr$ and $\alpha(s) = \int_a^s v(r) dr$. Since u, v are continuous and non-negative, α and β are continuously differentiable and increasing on $[a, b]$. This implies $\alpha \in \mathcal{R}(\beta)$ with $d\beta(t) = u(t)dt$ and $d\alpha(s) = v(s)ds$. Applying part (a) on the interval $[a, x]$ instead of $[a, b]$, the LHS becomes:

$$\int_a^x \left(\int_a^t f(s) d\alpha(s) \right) d\beta(t) = \int_a^x (\beta(x) - \beta(t))f(t) d\alpha(t).$$

Substituting the definitions of α and β back into the RHS:

$$\int_a^x \left(\int_t^x u(r) dr \right) f(t)v(t) dt = \int_a^x \left[v(t) \int_t^x u(r) dr \right] f(t) dt.$$

Thus, we define $w(x, t) = v(t) \int_t^x u(r) dr$, which is continuous in both variables.

If u, v are not non-negative, they are still bounded because they are continuous on a compact interval. Write $u = (u + M) - M$ and $v = (v + K) - K$ where $u + M \geq 0$ and $v + K \geq 0$. By linearity, expand the target equation into four integrals involving only non-negative continuous functions, apply the formula, and recombine to see that $w(x, t) = v(t) \int_t^x u(r) dr$ works universally.

Problem 6.11 (B:6.8:3 – Integrability of γ' is enough to prove the curve-length formula.). Suppose $\gamma : [a, b] \rightarrow \mathbb{R}^k$ is a differentiable curve such that $\gamma' \in \mathcal{R}$. Exercise **B:6.7:1(a)** tells us that given $\epsilon > 0$ we can find a partition $P = \{x_0, \dots, x_n\}$ such that

$$\sum_{i=1}^n \text{diam}(\mathbf{f}([x_{i-1}, x_i])) \Delta x_i < \epsilon.$$

- (a) Show that if P is such a partition, and we take any point t_i in each interval $[x_{i-1}, x_i]$, then

$$\sum |\gamma(x_i) - \gamma(x_{i-1})| - |\gamma'(t_i)| \Delta x_i < \epsilon.$$

- (b) Deduce that Theorem 6.27 remains true if the assumption " γ' is continuous" is weakened to " $\gamma' \in \mathcal{R}$ ".

(a) By the Fundamental Theorem of Calculus (Theorem 6.20) extended to vector-valued functions, we have:

$$\gamma(x_i) - \gamma(x_{i-1}) = \int_{x_{i-1}}^{x_i} \gamma'(t) dt.$$

Therefore, we can rewrite the term inside the absolute value:

$$\gamma(x_i) - \gamma(x_{i-1}) - \gamma'(t_i) \Delta x_i = \int_{x_{i-1}}^{x_i} (\gamma'(t) - \gamma'(t_i)) dt.$$

By the reverse triangle inequality $||A| - |B|| \leq |A - B|$, we establish:

$$||\gamma(x_i) - \gamma(x_{i-1})| - |\gamma'(t_i)| \Delta x_i| \leq |\gamma(x_i) - \gamma(x_{i-1}) - \gamma'(t_i) \Delta x_i|.$$

Using Theorem 6.12(e) to bring the norm inside the integral:

$$\left| \int_{x_{i-1}}^{x_i} (\gamma'(t) - \gamma'(t_i)) dt \right| \leq \int_{x_{i-1}}^{x_i} |\gamma'(t) - \gamma'(t_i)| dt.$$

Since $t, t_i \in [x_{i-1}, x_i]$, the distance between their images under γ' is bounded by the diameter of the image of the interval: $|\gamma'(t) - \gamma'(t_i)| \leq \text{diam}(\gamma'([x_{i-1}, x_i]))$. Thus, the integral is strictly bounded by $\text{diam}(\gamma'([x_{i-1}, x_i])) \Delta x_i$. Summing over all $i = 1 \dots n$ yields:

$$\sum_{i=1}^n ||\gamma(x_i) - \gamma(x_{i-1})| - |\gamma'(t_i)| \Delta x_i| \leq \sum_{i=1}^n \text{diam}(\gamma'([x_{i-1}, x_i])) \Delta x_i < \epsilon.$$

(b) Theorem 6.27 states that the length of the curve is $\Lambda = \int_a^b |\gamma'(t)| dt$. We must show this holds without assuming γ' is continuous.

First, for *any* partition P , we have:

$$\Lambda(P, \gamma) = \sum_{i=1}^n |\gamma(x_i) - \gamma(x_{i-1})| = \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt = \int_a^b |\gamma'(t)| dt.$$

Since $\Lambda = \sup \Lambda(P, \gamma)$, this shows $\Lambda \leq \int_a^b |\gamma'(t)| dt$.

To show the reverse inequality, let $\epsilon > 0$. Since $\gamma' \in \mathcal{R}$, Theorem 6.11 ensures $|\gamma'| \in \mathcal{R}$. Thus, there exists a partition P_1 such that $U(P_1, |\gamma'|) - L(P_1, |\gamma'|) < \epsilon$. We are also given a partition P_2 satisfying the diameter condition from part (a). Let P be a common refinement of P_1 and P_2 . The properties of both partitions hold for the refinement P . By part (a), for any choice of t_i :

$$\sum_{i=1}^n |\gamma(x_i) - \gamma(x_{i-1})| \geq \sum_{i=1}^n |\gamma'(t_i)| \Delta x_i - \epsilon.$$

Since P is a refinement of P_1 , the sum $\sum |\gamma'(t_i)| \Delta x_i$ is bounded between $L(P, |\gamma'|)$ and $U(P, |\gamma'|)$, which differ by less than ϵ . Consequently, the sum is within ϵ of $\int_a^b |\gamma'(t)| dt$. Thus, $\Lambda \geq \Lambda(P, \gamma) \geq \int_a^b |\gamma'(t)| dt - 2\epsilon$. Since ϵ is arbitrary, $\Lambda \geq \int_a^b |\gamma'(t)| dt$. Combined with the first inequality, $\Lambda = \int_a^b |\gamma'(t)| dt$.